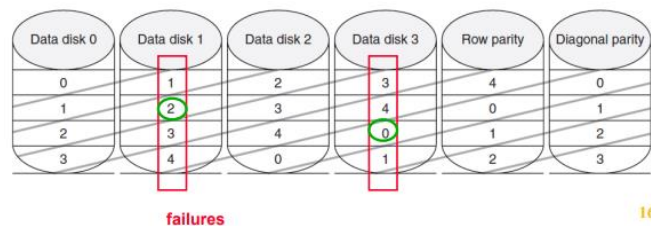


每題 10 分，多選不倒扣。

1. Suppose we would like to achieve a speed up of 5 with 10 processors, What fraction of the original computation can be sequential?
2. What are the advantages of PC relative conditional branch?
3. About Single-Cycle and Multi-Cycle implementation of CPU in computer organization, which of the following statements are NOT True?
 - A. Single Cycle implementation is used in the mainstream processors nowadays.
 - B. Single Cycle implementation of CPU, the clock cycle is determined by the shortest possible path.
 - C. Single Cycle implementation of CPU, allows a functional unit to be used more than once per instruction.
 - D. Compared to single-cycle implementation, multi-cycle implementation is more efficient.
 - E. Multicycle implementation is not suitable for pipeline implementation.
4. A. What protect mechanism that RAID 6 use? For a RAID 6 system illustrated in the figure below. If Disk 1 and the Disk 3 both fail at the same time, what would be the recovery order of all the data blocks?

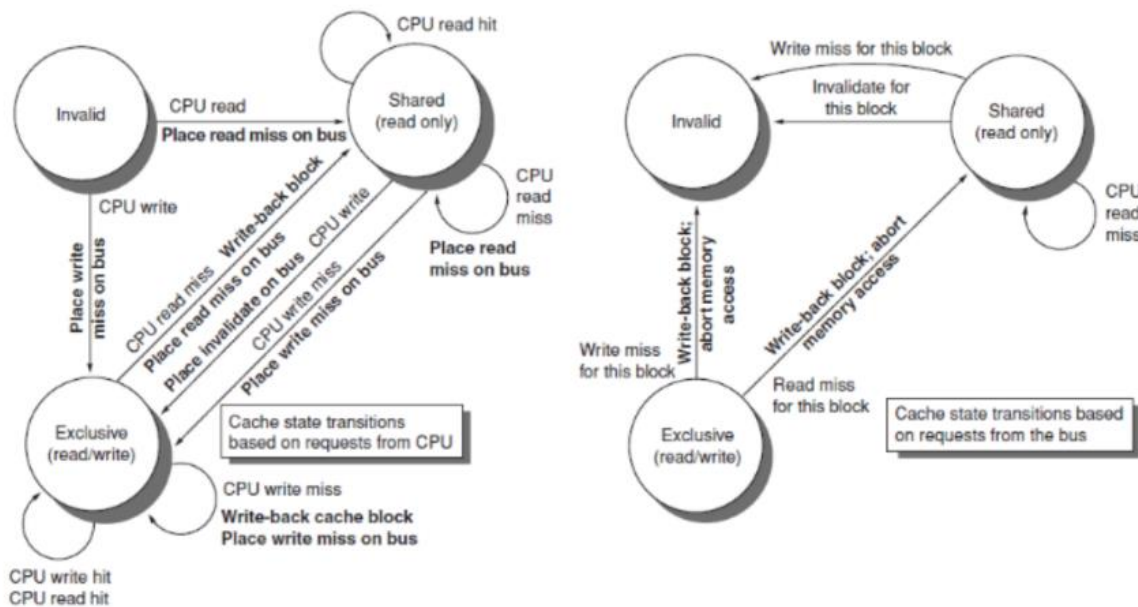


5. Suppose the length of the address is 32 bits. For a four-way set associative 512KB cache whose block is 64 Bytes, how many bits are used for index? How many bits are used for tag?
6. Which of the following statements are NOT true? (Multiple choice) (多選題)
 - A. For a cache with write back strategy, read misses might result in writes.
 - B. For a cache with write through strategy, a write buffer is not necessary.
 - C. Write after write (WAW) hazards can be resolved by register renaming.
 - D. Read after write (RAW) hazards can be resolved by register renaming.
 - E. Distributed shared-memory scheme will not result in non-uniform memory access time for multiprocessor machines.
7. Which of the following statements are true? (Multiple choice) (多選題)
 - A. Higher level of associativity can reduce capacity miss.
 - B. Higher level of associativity increases hit time.
 - C. Compiler optimization techniques cannot reduce conflict miss and capacity miss.
 - D. Adding a second-level cache can reduce miss rate.
 - E. Non-blocking cache can reduce miss rate.

8. The state transition diagrams of a write invalidate, cache coherence protocol for a write-back cache are illustrated below.

Suppose two processors P1 and P2 are both in Shared state for a data block X. P2 writes X at time T1. Then, P1 writes X at time T2. Afterwards, P2 reads X at time T3. Which of the following statements are true? (Multiple choice) (多選題)

- A. At time T1, P2 sends a CPU write request and place invalidate on bus and then goes to exclusive state.
- B. At time T1, P2 sees write miss for this block from the bus and goes to invalid state.
- C. At time T2, P1 sends a CPU write request and place write miss on bus and then goes to shared state.
- D. At time T2, P2 sees write miss for this block from the bus, it writes back the block, abort memory access and goes to invalid state.
- E. At time T3, P1 sees read miss for this block from the bus, it writes back the block, abort memory access and goes to shared state.



9. Merging Arrays and Loop Interchange is the way to reducing Misses by Compiler Optimizations, please give example of each way and explain why they can reduce misses?

10. We have ideal CPI is 2.0, and 1.3 memory references per instruction. And we have Miss rate 3% and Miss Penalty 60 cycles.

Calculate the effective CPI considering the cache miss rate and miss penalty.

答案：

1. $5 = \frac{1}{x + \frac{1-x}{10}}$, $x = \frac{1}{9}$

2.

- branch address is specified by an address field that is added to Program Counter
- The target address is often near the current instruction.
- Because the absolute address is too large, it can be more flexible.

3. ABCE

4. Network Appliance's row-diagonal parity.

先看對角，因為優先找對角線只剩餘 1 個需要被 recover 的，因此 recover Disk1 的 2 跟 Disk 3 的 0

再看 recover 後的列，因為第二列跟第三列只有 3, 4 剩餘一個要 recover，因此 recover Disk1 的 3 跟 Disk 3 的 4

再看一次對角，因為所有對角線只有 3, 4 只剩餘一個要 recover，因此 recover Disk1 的 4 跟 Disk 3 的 3

再看 recover 後的列，也只剩下第一列跟第四列只有兩個 1 要 recover，因此 recover Disk1 的 1 跟 Disk 3 的 1

5.

Byte offset : $64 = 2^6 \Rightarrow$ byte offset is 6 bits.

Index : $2^{\text{index}} = \frac{512 \times 2^{10}}{64 \times 4} \Rightarrow$ index bits = 11 bits.

Tag : $32 - 11 - 6 = 15$ bits.

6. BDE

7. B

Ps. A 似乎課本沒提及，筆者認為不能選。

8. ADE

9.

Merging Arrays Example

```
/* Before: 2 sequential arrays */
int val[SIZE];
int key[SIZE];

/* After: 1 array of structures */
struct merge {
    int val;
    int key;
};
struct merge merged_array[SIZE];
```

- Some programs reference multiple arrays in the same dimension with the same indices. (these accesses will interfere with each other)
- Reducing conflicts by combining independent matrices into a single compound array so that each cache block can contain the desired elements; improve spatial locality

Loop Interchange Example

```
/* Before */
for (k = 0; k < 100; k = k+1)
    for (j = 0; j < 100; j = j+1)
        for (i = 0; i < 5000; i = i+1)
            x[i][j] = 2 * x[i][j];

/* After */
for (k = 0; k < 100; k = k+1)
    for (i = 0; i < 5000; i = i+1)
        for (j = 0; j < 100; j = j+1)
            x[i][j] = 2 * x[i][j];
```

Row major
or column
major

- Sequential accesses instead of striding through memory every 100 words; improved spatial locality

10. $2 + 1.3 \times 3\% \times 60 = 4.34$